Accessing and Improving Energy Efficiency of Rural Piped Water Supply Systems

by

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Abstract

Piped Water Supply Systems (PWS systems) are massive consumers of energy. Energy is consumed at various stages of the scheme from pumping the water from the source, transmission over a distance, distribution to various consumers or overhead tanks if any. Pumping accounts for almost as high as 80%of the total energy cost and also a significant part of overall cost of operation and maintenance of water works. The scheme documents currently do not take into account the quality of supply and reliability into consideration while designing the pump control hence leading to inflated energy bills, break down of equipment and hence leading to failure of schemes. There are also considerable losses in the water produced and distributed because of the leakage in the pipelines, unauthorized service connection etc. The above mentioned causes coupled with lack of proper piping lay out, use of substandard pipes, valves, pumps and also lack of sound O&M practices leads to lot of schemes going defunct. The aim of this project is to create a sample template to assess the energy efficiency of the PWS schemes and test the template on schemes in real time to understand the bottle necks involved and suggest possible recommendations to be included in the design documents. This project envisages in understanding the PWS scheme structure in Shahpur Taluka, Thane district, Maharashtra.

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List of abbreviations

Abbrevation	Full Form
PWS	Piped Water Supply systems
NRDWP	National Rural Drinking Water Program
TDH	Total Dynamic Head
MJP	Maharashtra Jeevan Pradhikaran
LPCD	Litres Per Capita Per Day
MSEB	Maharashtra State Electricity Board
MSEDCL	Maharashtra State Electricity Distribution Company Ltd
MBR	Mass Balancing Reservoir
ESR	Elevated Storage Reservoir
B.E.P	Best Efficiency Point
O&M	Operation and Maintenance

1. Introduction

Under National Rural Drinking Water Program (2009) the Ministry of Drinking water and sanitation has set out a framework for planning, implementation and management of drinking water schemes to ensure drinking water security, institutional and financial support and capacity building. The strategic plan of 2011-2012 aims at providing drinking water via Piped Water Supply to 55% [1]of the households and house hold connection to 35% of the households by 2017. In light of this ambitious project it has been observed that many schemes go defunct due to high energy bills and other operational issues. It is wise to see how the energy parameter influences the scheme's life and its functioning.

1.1 Background

The integral relationship between water and energy is often not understood and dealt with systematically so as to address the efficiency of the system. The energy consumption of the different steps that are needed to transport water from source to destination depends on factors like the topography between the water source and its destination (especially elevation change), distance from the bulk water supply, and the quality of the primary mains (supply pipes) and secondary mains distribution pipes). Globally, energy is among the top three cost items to water utilities, in developing countries, energy is usually the highest cost associated with water supply. Water Supply systems are also capital intensive in nature requiring construction of pump stations, conveyance mains and distribution channels. Water access can be expanded much more quickly and inexpensively through efficiency than new infrastructure. Thus, by incorporating efficiency into design the costs can be controlled, the service improved and also access can be expanded.

A few clauses which relate to energy efficiency are also included in the D.P.R (Detailed Project Report) for preparation of water supply system and also MJP Clauses addressing the same. The D.P.R requires the following[2]:

(i) Availability of three phase power in hours/day

(ii) Type of pump used–submersible or solar driven or combination of both

(iii) Total Dynamic head (TDH) and LPD

The MJP technical circular dated May 23, 2002 mentions that during the time of preparation of plan/scheme or even while tendering the realistic number of hours of supply that can be possibly provided should be enquired from M.S.E.B as well as local public and accordingly the pumping system be designed. In spite of the above mentioned clauses being included in the documents the experience from the field work carried out proves otherwise. The pump station operators are clueless about the number of hours of electricity that is provided as the supply is erratic, especially during the summers where there is a huge demand supply gap. Also, the design documents do not consider the quality of supply which is supplied to the pump stations, and can have serious ramifications on the performance of the pump, hence affecting its efficiency. The Total dynamic head (TDH) and (LPD) are taken note of in the design phase, but the extent to which they can vary safely so that the system operates at least within 20-30% of its B.E.P (Best Efficiency Point) is not taken care of.

1.2 Motivation

Energy-water nexus has been a concern for quite some time due to the depleting aquifers since the ever increasing cost of energizing alternatives it is a good idea that we observe some case studies. The energy-water nexus problem was addressed to some degree so that we have an idea of the alternatives, strategies and problems associated with managing this nexus.

Water Supply Systems are massive consumers of energy since energy is needed right from sourcing, conveyance to treatment and distribution provided if it is not gravity fed. Pumping has a major share of the energy close to 80%[3]. The ever widening energy-water nexus mandates us to look into the aspect of energy efficiency of water supply schemes in detail so as to avoid the failure of schemes due to inefficient operation as well as developing scheme design documents which also include an energy section which predicts the possible energy demand of the scheme over its life time so as to make the utilities ready for eventualities.

The primary reason for the failure of the PWS is the burgeoning cost of the energy bills which are defaulted and hence leading to disconnection of the service. The inflated energy bills are the result of inefficient pumping, operating pumps below the best efficiency point, dry running of the pumps, starting the pump with delivery valve open leading to burn outs and operating at a head higher or lower than actually needed.

Improper design of piping and excessive pressure at nights due to faulty Load scheduling leads to leakage of water through pipes, joints and glands of sluice valves. Unreliability of supply to pumps leads to rapid closure of valves which in turn leads to a surge in the pressure causing water hammer, which may have a sufficient magnitude to disrupt the transmission pipe or damage the valves fixed on the pipe. The problems in distribution systems arise due to unscheduled demand leading to peak demand hence affecting the pressure. Maintenance of appropriate pressure at all supply points is the main concern of O&M. Aging of the pipes leads to reduction in the carrying capacity due to incrustation, hence it is a logical conclusion to study the above factors in addressing efficiency of the scheme.

2. Essential components of a PWS scheme

2.1 General block diagram of a Public Water Supply Pumping scheme.

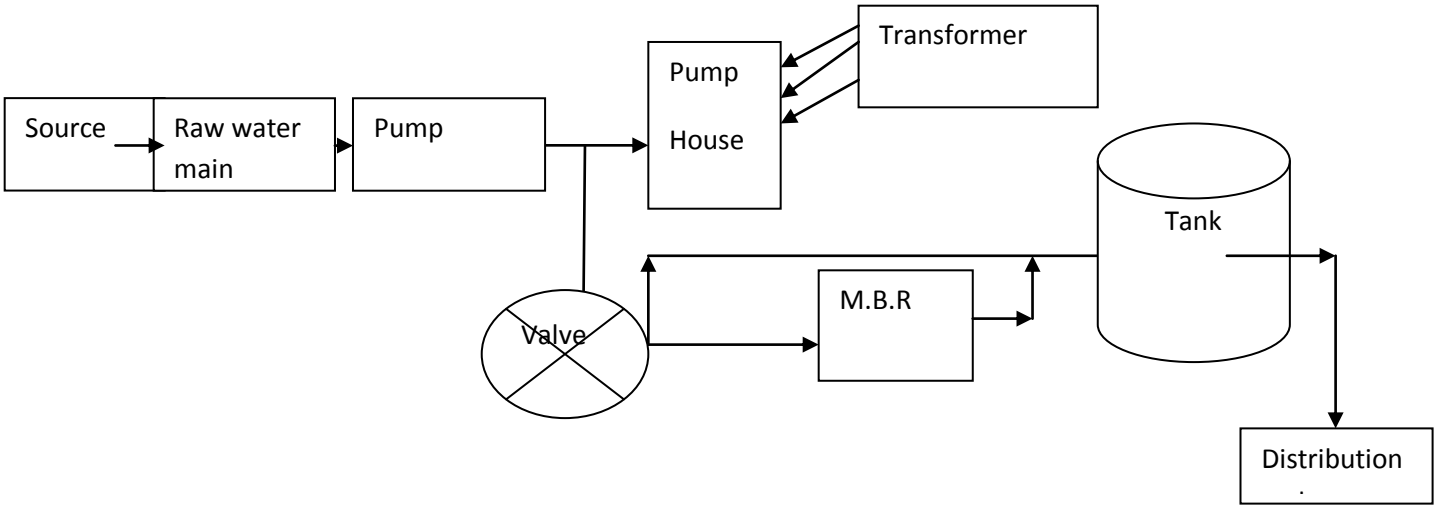


Figure 1: Block diagram of a Piped Water Supply Scheme

Description of the components:

(i)Source: The source can be an open well, closed well or surface water source.

(ii)Pump: The Pump could be a centrifugal, mono block, positive displacement multi or single stage according to the requirement. Solar or solar hybrid can also be used. Generally 3 phase A.C pumps are used as they are robust.

(iii) Pump House: The pump house consists of the pump machine room housing the Motor control Centre. Power factor correction equipment, other power measuring devices if any, Valve control mechanism and the metering for the scheme.

The M.C.C Panel houses the motor starter which acts as on/off mechanism for the pump.

(iv)Transformer: The transformer supplies power the pump house from the L.T line typically it is 3 phase, oil cooled and has KVA rating on the nameplate mentioned.

(v) Mass Balancing Reservoir: Few schemes may consist of MBR which store the water before releasing upstream to aid in scheduling.

(vi)Tank: an overhead tank collects water from the system and gravity feeds to the distribution system down side.

(Vii) The distribution network entails secondary and tertiary network from tank which supply water to the customer end the pipes used are made of different types of material like Galvanized Iron, PVC, Cement etc.

2.2 Pumping System and Parameters

The amount of energy consumed in a P.W.S system mainly depends on pumping equipment which in turn dictate the head and flow rate and to some extent on other parameters like time of the day tariffs, diameter of the pipe used, operating hours etc. Since pumping is the primordial determinant of any P.W.S scheme's energy consumption, it is a good idea to look at the pump characteristics and the factors that affect the energy consumption. In some cases, pumping system energy is used quite efficiently; in others, it is not. Pump operators are often very familiar with the controllability (opening/closing of valves), reliability (maintenance issues), and availability of pumping system equipment, but they might not be aware of system efficiency which is very important. There is a strong correlation between the reliability of pumps and their efficiency; Pumps that operate close to Best Efficiency Points tend to perform more reliably and with greater efficiency[4].

Given the significant amount of electricity attributed to pumping systems, even small improvements in pumping efficiency could yield very significant savings of electricity. The pump is among the most important of the components that comprise a piped water system, including the motor, transmission drive, piping and valves. It is a good idea to consider few pump parameters before delving into the efficiency issues of the system.

Pump Parameters

(i) Discharge Head: Discharge head is simply the difference in height of the supply and destination reservoirs It is independent of the flow and remains constant.

(ii) **Frictional Head:** Friction head (sometimes called Dynamic Head Loss) is the friction loss, on the being moved, in pipes, valves and equipment in the system. Friction tables are universally available for various pipe fittings and valves. The friction losses are proportional to the square of the flow rate. It also depends on the length of the pipe.

(iii) **Flow Rate:** The rate at which the liquid is pumped by the system is called Flow Rate.

(iv) **Suction Head:** Suction head can be positive or negative. It is the distance between water level and centreline of the suction side of the pump. If the water level is above the line, it is negative and if it is below it is positive.

(v) **Total Head**: Total head is a sum of static, suction and friction heads.

Pump Operating Point

The performance of a pump can be expressed graphically as head against flow rate. The centrifugal pump has a curve where the head falls gradually with increasing flow. This is called the Pump Characteristic Curve (Head – Flow curve). The system curve is a combination of static and frictional head which intersects the pump curve at some point and it is called as pump operating point.

If the actual system curve is different to the one which is calculated or if the system curve varies over a period of time due to variation in the hydraulic profile because of seasonal variations or increased frictional losses due to encrustation in the pipes, clogging of the valves, clogging of strainer gauge etc the operating point of the system changes and the pump will operate at a different operating point than expected. In general a safety margin is considered and a pump with a larger capacity is used which operates at a higher flow rate orin a throttled condition and hence leads to inefficient operation.

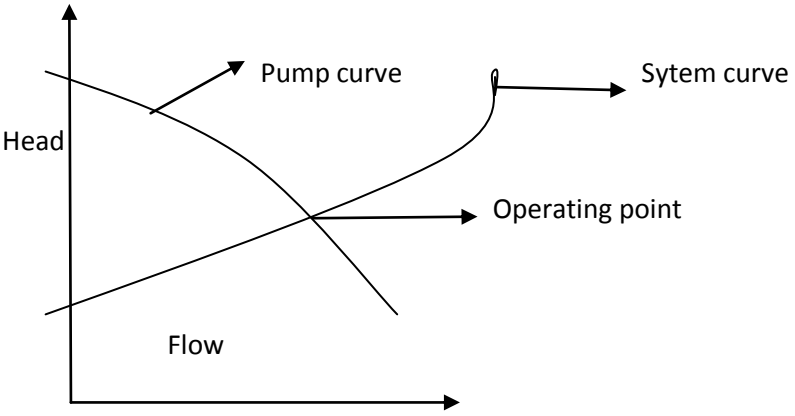


Figure 2: Head v/s flow curve showing operating point in a pump

2.3 Prime Movers

Most pumps are driven by electric motors. Although some pumps are driven by direct current (dc) motors, the low cost and high reliability of alternating current (ac) motors make them the most common type of pump prime mover. The efficiency of motor plays an important role in determining the overall efficiency of the system as it is cascaded with the pump and the overall efficiency of the system is the product of both pump and motor efficiencies.

2.4 Motor Control Centre

A subcomponent of a pump motor is the motor controller. The motor controller is the switchgear that receives signals from low-power circuits, such as an on–off switch, and connects or disconnects the high-power circuits to the primary power supply from the motor.

2.5 Pipes

The critical aspects of piping are its dimensions, material type, and cost. Since all three aspects are inter related, pipe sizing is an iterative process. As the diameter of the pipe increases the flow rate increase; however, larger pipes are heavier, take up more floor space, and cost more than smaller pipe. Small diameter pipes restrict flow. They also operate at higher liquid velocity, increasing erosion effects, wear, and friction head. Increased friction head affects the energy required for pumping as the pressure builds up to deliver the same amount of liquid at a higher head[5].

2.6 Valves

The flow in a pumping system may be controlled by valves. Some valves have distinct positions, either shut or open, while others can be used to throttle flow. There are various kinds of valves and selecting one to application specific requirement is a matter of frequency of use and reliability.

Valves can be used to isolate equipment or regulate flow. Isolation valves seal off a portion for maintenance whereas Flow-regulating valves either restrict flow through a system branch (throttle valve) or allow flow around it (bypass valve). A throttle valve controls flow by increasing or decreasing the flow resistance across it. In contrast, a bypass valve allows flow to go around a system component by increasing or decreasing the flow resistance in a bypass line. A check valve allows fluid to move in only one direction, and prevents back pressure.

3. PWS Schemes Energy Data Analysis

To gain an insight into energy consumption and billing of PWS schemes, a taluka was chosen from Maharashtra. The individual scheme related energy and billing data was obtained from MSEDCL and a brief analysis is carried out.

The current analysis gives us an idea about the patterns in energy consumption. This could be useful in disbursing subsidy if any being planned by NRDWP, as it gives an idea about the quantum of consumers needing subsidy. Various parameters like subsidy amount/kWh consumed, or kWh slab for subsidy can be decided.

Shahpur Taluka is chosen for the purpose of study, a brief outline about the Taluka is given below:

N.o villages:220 villages

N.o habitations: 662

Population: 292000

The working schemes have variations in terms of energy consumed, the design of each scheme varies based on the local conditions like total dynamic head, pumping system used, source type, pipe lay out etc. There are over 600 PWS schemes in entire Shahpur taluka, the data obtained from MSEDCL included the bill amount and the unit consumption for the schemes over a period of 15 months.

3.1 Frequency Distribution of the schemes according to the number of units consumed

As expected, the distribution is normal in nature with the highest number of schemes lying in the band width of 10000-15000 units.

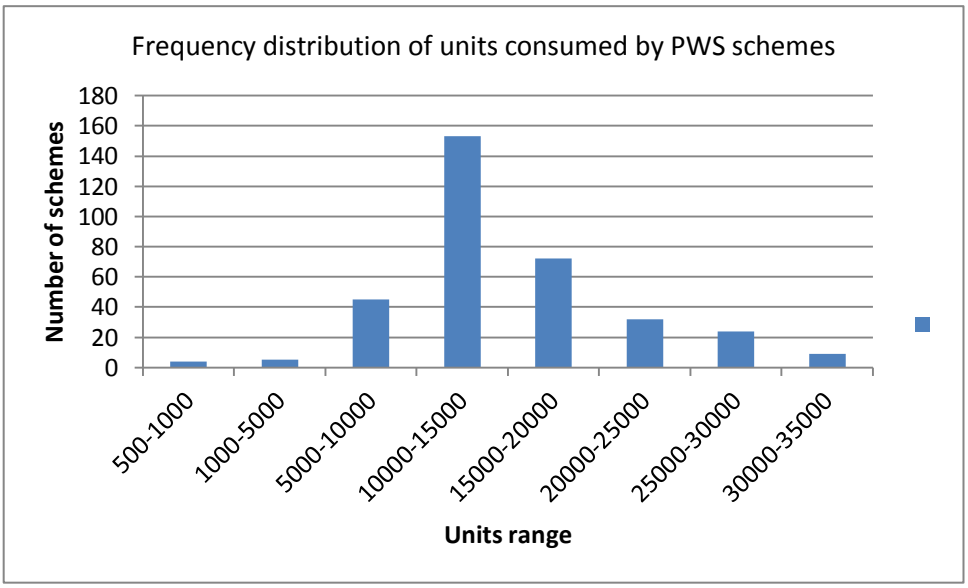


Figure 3: Frequency distribution of the number of schemes v/s units

3.2 Deficit Bill amount:

To test the payment rate of the PWS schemes 15 month data about the actual bill and bill payed is calculated for about 100 schemes and the difference between both of them is evaluated. Most of the schemes are running in deficit, which clearly indicates that the payment health of the schemes is bad. If the schemes continue to not pay up, they could go defunct.

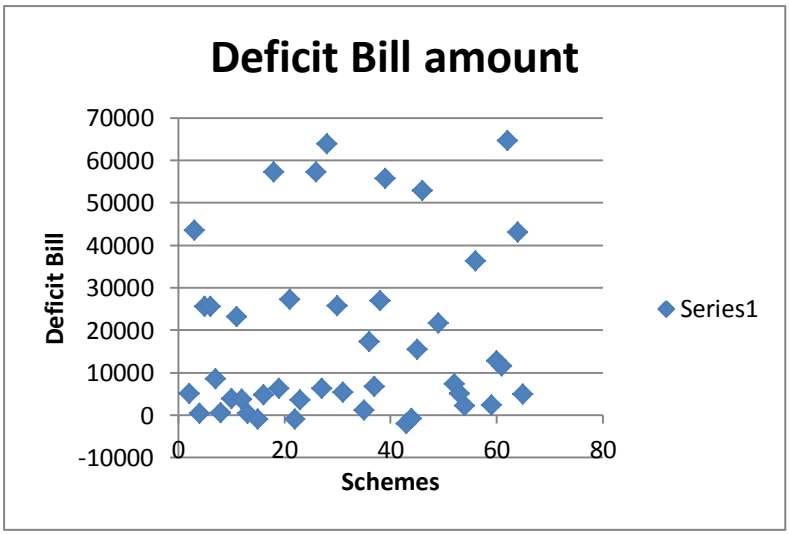


Figure 4: Deficit bill amount of PWS schemes

4. Energy Efficiency analysis of a PWS scheme

4.1 Problem statement and necessity to determine the energy efficiency of a scheme:

As discussed earlier, the energy consumption of a PWS scheme is one of the important parameters in defining the scheme's viability to survive. Hence, determining the energy consumption and predicting the possible changes in consumption over a period of time is essential to have an idea about the costs that will be incurred such that the schemes do not go defunct due to inability in payment of the energy bills.

The primary energy consuming entity in a Piped Water Supply Scheme is the pumping system. When the pumping system is designed for a particular head and flow rate, the design point specifies a particular efficiency of operation. It is called Best Efficiency Point, and pumps are designed to operate within the B.E.P. region[6]. Based on the rating of the pump the B.E.P efficiency varies between 65%-80%[7]according to IS 14220:1994the standard for submersible pumps. Hence before installation of the pump one can have a rough idea about the energy that is going to be consumed to deliver water at a particular flow rate, at a particular head for a fixed number of hours.

The energy bills of the scheme give an idea of the actual energy that is being consumed. If the energy that is being consumed is close to one that is predicted, the scheme seems to be running well. If there is a considerable variation in the schemes actual energy consumption that means the scheme isn't running efficiently. The possible reasons for the inefficient operation of the scheme could be one or more of these as mentioned below.

(i) Motor efficiency
(ii) Pump efficiency
(iii) Foot valve type
(iv) Pipe lay out and dimensions
(v) Voltage imbalance
(vi) Harmonics
(vii) Pipe leakages
(viii) Over Heating of the motor

Depending on the type and location of the pump and available design documents which can be procured from the concerned water supply office in the Taluka, one can determine the effect of all or some these parameters on the scheme functioning and hence see if the scheme is operating efficiently according to the design . Here, an attempt is made to create a sample template to address the energy efficiency of the scheme. This template can be used on any scheme by following the guidelines indicated. We can assess how much the energy consumption is overshooting the expected energy consumption of the scheme.

4.1 Sample template for calculating the Energy Efficiency of the Scheme:

i) Pre-field details

Before proceeding to the field to assess the energy efficiency of the scheme, it is necessary and advisable to keep the below. Procure the scheme design documents from the Maharashtra Jeevan Pradhikaran office from the Taluka in which the scheme is located and check for the below details.

Table 1: Sample Template for Energy efficiency -Pre-field details

Name	Comments
Capacity of the scheme	In H.P/kW
Number and type of pumps installed	Submersible/Centrifugal/Stand by pumps if any
Suction head (in meters)	
Discharge head (in meters)	
Number of pumping hours the system is designed for	
Capacity of the transformer (KVA)	
Energy bills of the scheme for a period of 6-12 months	This is needed in order to estimate the amount of energy that is being consumed actually and compare with the energy bills procured.
Basic idea/blue print of the scheme	It is necessary to have a blue print of the scheme before visiting the scheme as it would guide you to explore different aspects of the scheme like Jack well, Sump well, MBR , over head tank locations and pipe diversions if any so as to have an idea about how the head varies or if there are any leakages along the pipeline.
Power Monitoring unit	Power Monitoring unit is needed to measure the input electrical energy being consumed by the (motor+pump) unit. A detailed circuit diagram with connections and wiring will be attached as an appendix.

Data Logger	Data logger is used to record parameters like (Kwh, KVAr, KVA, Power factor, THD%, Voltage in three lines, Current in three lines) over a specified period of time.
Flow rate measurement/Ultrasonic Flow meter(UFM)	Determining the flow rate of the pump is necessary for measuring the output power if the pump cannot be bolted out use a UFM. Else look for an auxiliary supply or any other method to measure the flow rate. Keep in mind that the flow rate should be measured as close to the pump as possible since the farther you go there is a possibility of error in the flow rate as there may be leakages or the pipe diameter might vary hence changing the flow.
GPS device	Useful in measuring the pressure head.
Screw drivers, Tester, wires,leads,connectors,tapes, Multimeter	Useful in connecting the equipment to the motor

II) On-field measurements

Once reaching the pump house the following parameters should be measured/ observed to calculate the efficiency.

Rig up the Power monitoring unit, current transformers and data logger and measure/log the following. Keep in mind to switch off the Fuse and Breakers before attempting measurements.

Table 2: On-field Measurements

Name	Comments
Voltage in three lines (R,Y,B)	To see if the voltages are balanced and at rated level for a three phase supply.
Current in three lines	To understand the over current, under current drawl due to winding defects if any in the motor.
KWh consumed	Used for Efficiency calculation as input parameter and also compare with energy bills procured earlier.
Kvar consumed	To understand the reactive power being consumed requirements and accordingly plan power factor correction if needed using capacitors.
THD(%) total harmonic distortion	Measure the THD(%) to see if the harmonics in the system are well within the limits, else inline harmonic filters are needed to be installed.
Power interruption timing	To calculate power continuity index and hence estimate if the power availability is according to the claimed availability of power in the design documents.
Discharge head	Use GPS device to measure the discharge head between source and final destination
Suction head	(Total depth)-(depth at which pump is placed)-(water level from the top)
Flow rate (Q)	Determine the flow rate of the water at the pump discharge end by using an Ultrasonic Flow Meter or by any other auxiliary methods if convenient.
Pump name plate ratings and part number	Useful in gathering information about pump characteristic curves from the manufacturer to determine Best Efficiency Point.

Pipe diameter, material and lengths	Useful in determining friction coefficients so as to calculate frictional head
Major leakages	Observe for major leakages if any in the piping and if possible determine the rate of leakage, this information could be useful in estimating loss factor

4.2 Common Pump Parameters and Block diagram

We are primarily interested in calculating the efficiency of the (pump+motor) unit, hence we need to determine the output hydraulic power at discharge end of the pump and input electrical power.

Since the exact discharge end of the pump is not accessible in submersible systems, we calculate the hydraulic power elsewhere as close to the pump as possible and compensate accordingly by adding the (static and friction head) to the power calculation. In a submersible type, it is not possible to determine the individual efficiencies of motor and pump unit though it would be ideal and advisable to do so. Few parameters that we frequently encounter during our discussion on energy efficiency and their definitions are given below.

Design Head: Static head (Includes suction head and discharge head) + estimated friction head during design phase.

Design Flow Rate: Estimated flow rate according to the requirement i.e. the number of hours needed to fill a tank and or the number of interconnected tanks connected to the system.

Measured Flow Rate: The flow rate ideally should be determined right at the discharge end of the pump since we need the output hydraulic power at the pump end to calculate efficiency of the pump. But if the pipe diameter doesn't change and if there are no leakages from the point of discharge of the pump to the measuring point, the flow rate can be determined at another point.

Measured Pressure: The pressure gauge should be used to determine the pressure at the discharge end of the pipe in order to measure. Now since the discharge end is not accessible, measure the pressure at any other point and add the static head (suction as well as discharge) as well as friction head by calculating the height as well as the length of the pipe from the discharge end of the pipe to that point, so in effect it is equivalent to measuring the pressure at the pump discharge end.

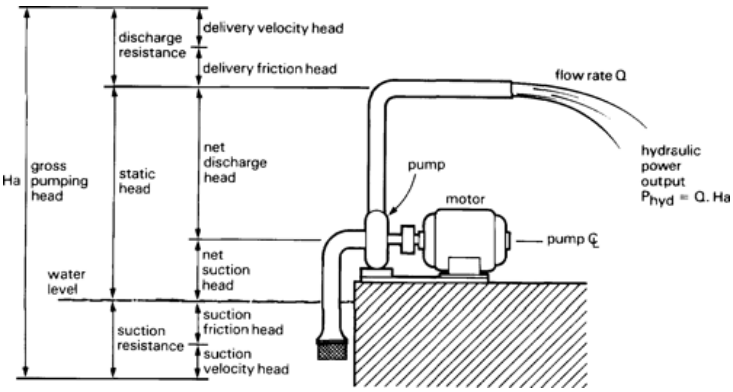


Figure 5: Block diagram of the pump showing different parameters[8]

A few things needed to calculate efficiency:

(i) The total depth of the well/sump (determine from design document)

(ii) The depth at which the pump is placed (design document)

(iii) The suction head (depth at which pump is placed – depth of the water level)

(iv) Impeller diameter

(v) RPM of the pump

(vi) Pump performance curves from the manufacturer

4.3 Methodology

We are primarily concerned with the (pump and motor) efficiency as it is the heart of a pumping scheme. Since, the pump is submersible (in most cases) and can't be individually tested for motor and pump efficiencies, we test the unit as a whole and then determine the actual pump efficiency from the pump manufacturer curves at the current duty point of pressure and flow rate, from this we can determine the actual motor efficiency . This gives us an idea about the extent to which both pump and motor might be deviating from their design efficiencies.

The following steps are followed in accessing the efficiency of the system

(i) Calculate the efficiency of the (pump+motor) theoretically, by measuring the input electrical power being consumed and output hydraulic power being delivered by the pump.

(ii) Input electrical power is measured by the power monitoring unit at all the times. We measure the flow rate and Pressure at different intervals of time and note down the corresponding power reading.

(iii) Convert the pressure measured at the gauge (at some other point) into equivalent head and to that add the static head from the pump end and also friction head

(iv)Measure the flow rate in the pipe using an Ultrasonic Flow Meter.

Evaluate Hydraulic power Ph (Qpgh/3.6*10^6) kW

Q: flow rate in m^3/hour

P: Density of water kg/m^3

g: acceleration due to gravity in kg/m^2

h: is the total head upto the point of measurement .

V1,V2,V3: Voltage in the three lines

P kW: Active Power consumed

P.F: Power factor

Efficiency of (Pump Motor) system is ζ total =Output hydraulic power / Input Electrical Power.

But total efficiency is a product of individual efficiencies of motor and pumpi.e

ζtotal = ζmotoractual* ζpump actual

ζtotal: Efficiency of (pump+motor) unit ; ζmotor actual: ActualEfficiency of operation of motor

ζpump actual : Actual Efficiency of operation of pump

Ideally, one has to determine motor efficiency by conducting the blocked rotor and Open circuit tests. Since the pump is submersible, conducting the tests on the motor unit may not be possible unless it is disconnected and taken out which is not practical in most of the cases.

(ζmotordesign*ζpump design)/(ζmotor actual*ζpump actual) = Pactual (kW)/ Pdesign(kW)

If operation is considered for a fixed number of hours per month the ratio is equal to Energy consumed per month (kWh or units) / Energy expected to be consumed per month.

ζmotor design and ζpump design are available according to IS standards (IS 14220:1994 for pump) and (IS 9283: 2013 for motor) . ζtotal (ζpump actual *ζmotor actual) is calculated from the measurements.

If the above calculated ratio above is close to 1, the scheme is operating according to expectation. If it is much higher than 1, then the scheme is consuming more energy than intended to. The cause of which can be multiple in nature.

The B.E.P efficiency of the pump is mentioned by the manufacturer at a particular speed and for a particular frequency, so any deviation from the speed or frequency will change the B.E.P. Hence ideally one has to measure the shaft speed and determine the change in rpm or frequency if any and accordingly correct the B.E.P efficiency point.

5. Case Studies and testing of template.

The sample template to measure efficiency of a PWS scheme has been discussed followed by energy efficiency calculation methods, in this section a few schemes have been considered where the sample template can be tested.

5.1Khardi Piped Water Supply Scheme:

It was designed in 1983-84 by MJP, the then WSSB(Water Supply and Sewerage Board) to supply piped drinking water to Khardi and five other villages namely Umbarkhand, Chanda, Golban, Lahe and Kukambe in Shahapur taluka of Thane district in Maharashtra. Currently, only three villages namely Khardi, Lahe and Kukambhe are serviced throughout year. The remaining three villages namely Umbarkhand, Chanda, Golban are supplied water from the scheme only during pre-summer and summer months (January to May).The scheme is based on tail water of Bhatsa dam as its source.

Population targeted is around 8000, but the population is growing due to the real estate boom in the surrounding regions hence placing more stress on the supply system.

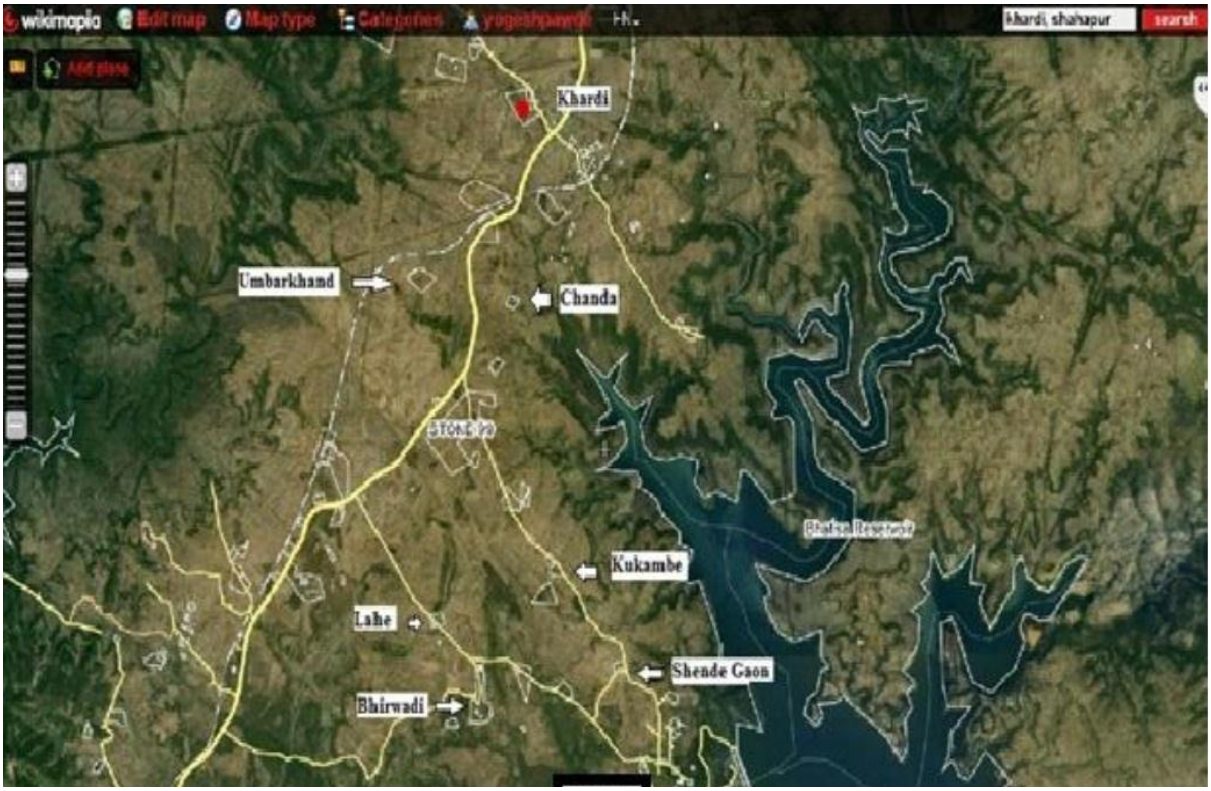
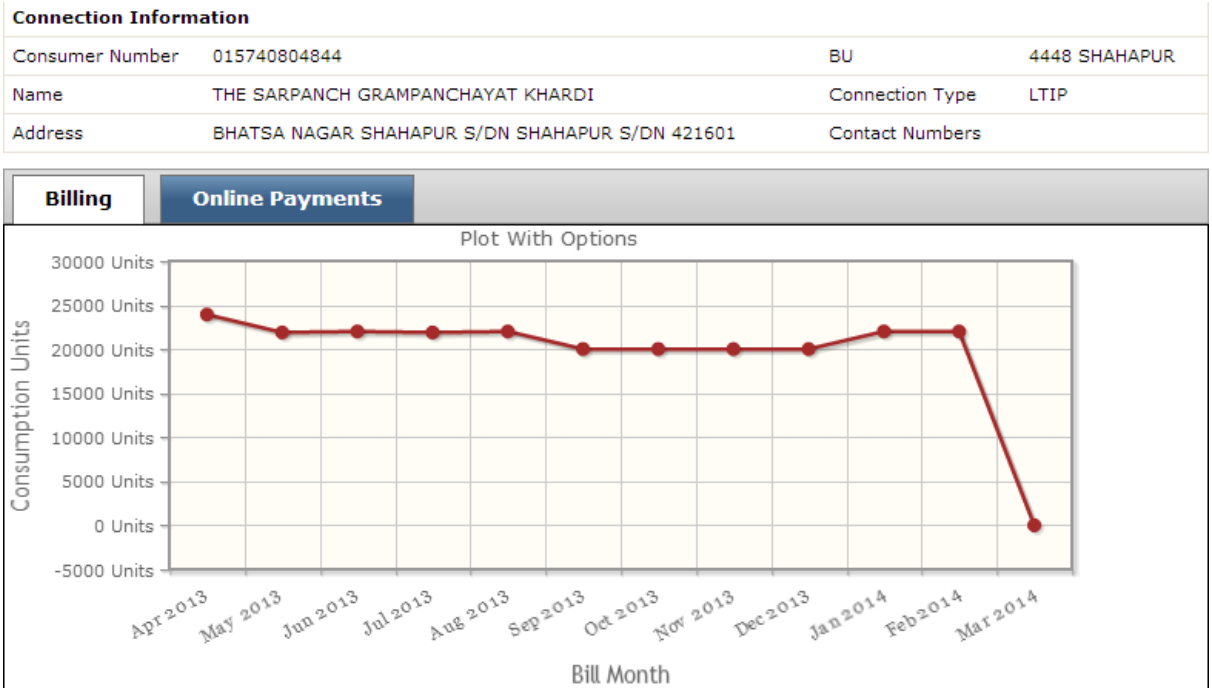


Figure 6: Map of Khardi Piped Scheme showing the beneficiary villages

5.2Energy consumption of the scheme

The scheme is a two stage pumping system. The water from the Bhatsa dam is initially collected in a jack well through raw water main and then is pumped into sump well from where it is pumped to the villages with a pipe line of around 11 k.m which has many leakages and maintenance is tough. The pipe line from jack well to sump well doesn't have any major leakages and since the distance is just 1.2km it can be maintained easily. From the consumer index numbers obtained from MSEDCL the unit consumption of both pump houses for different months is known, using a power monitoring unit and flow meter the power and flow measurements were carried in both pump houses to calculate the efficiency of the scheme and measure other parameters.



Connection Information

Consumer Number	015740804844	BU	4448 SHAHAPUR
Name	THE SARPANCH GRAMPANCHAYAT KHARDI	Connection Type	LTIP
Address	BHATSA NAGAR SHAHAPUR S/DN SHAHAPUR S/DN 421601	Contact Numbers	

Figure 7: Unit consumption of sump well /month

Table 3: Sump well details

Parameter	Value
Dimensions of the sump	9m*5m*3.3m
Design head	180m
Design flow rate	10lit/sec or 36m^3/hour
Level at which pump is placed inside the sump	2 m
Design efficiency from pump curve	65%

Table 4: Measured Parameters at Khardi

Parameter	Value
Pressure at the pressure guage in metrers	11.2 kg/cm^2 or 113 meters
Discharge head till the guage	2metres
Friction head till the guage	0.02m
Suction head	1.3m
Measured flow rate	40 m^3/hour
Pump efficiency for the given duty point	45% (approximately)

Table 5: Voltage and Power and Power factor readings -Khardi

The Voltage and power readings are average readings over a period of three hours, taken every five minutes.

V1	V2	V3	P(kW) total	P.F
229.5	230.2	234.1	35.2	0.91

5.3 Power and efficiency calculations

Output Hydraulic power: Q*ρ*g*h : 12.64 kW

Q: flow rate = 11.2 lit/sec or 40.32 m^3/hour

h : (Pressure at the pressure guage in metrers)+(discharge head till the guage)+(friction head till the guage)-(suction head)

113m+2m+0.02-1.3=113.72m

ρ: Density of water =1000kg/m^3

g: acceleration due to gravity =9.8kg/cm^2

Table 6: Power and efficiency readings-Khardi

Flow rate(m3/hour)	Pressure (kg/cm^2)	Output hydraulic power (kW)	Input Electrical Power (kW)	TotalEfficiency (ζactualpump*ζactual motor)
43.2	11.5	12.64	36.4	0.34
43.2	11.5	12.64	36.23	0.34
43.2	11.5	12.64	35.4	0.35
43.4	11.5	12.64	35.2	0.36
43.4	11.5	12.64	36.1	0.35
43.4	11.5	12.64	35.32	0.35
44.2	11.5	13.12	36.21	0.36
43.2	11.3	11.98	35.21	0.34

ζtotal =ζpumpactual*ζmotor actual

Hence , now the ζmotoractual = ζtotal/ζpump actual

= 77%

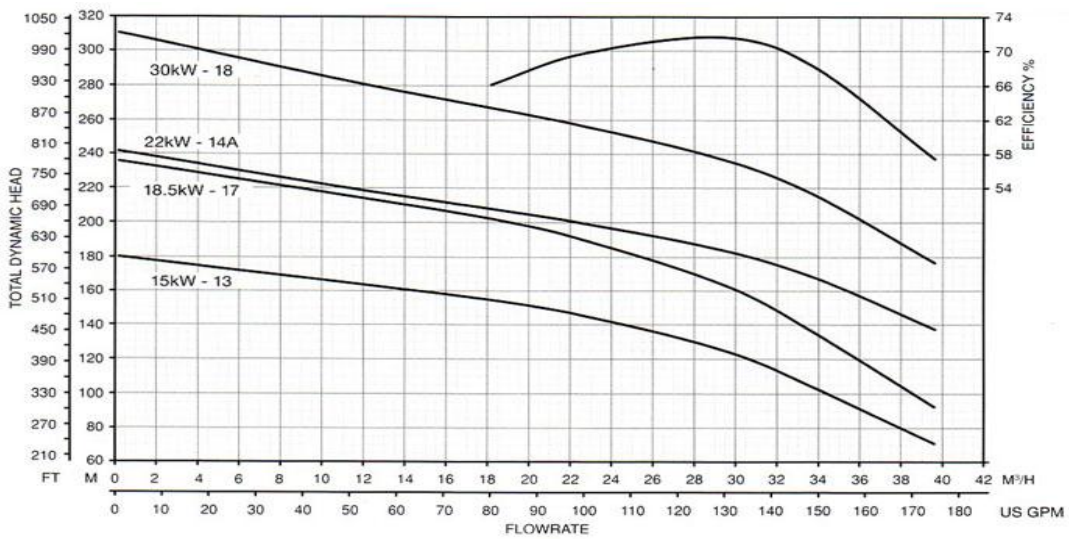


Figure 8: Pump characteristic curve for 30kW, 18A, Submersible pump[3]

5.4Comparing Expected and actual Power consumption:

The input electrical energy consumed is a function of both pump and motor efficiency for a given output power. According to the design efficiency of motor and pump the input power consumed is expected before hand, considering a little deviation we will have an idea about the power that will be consumed. Now, actual measurements of motor and pump efficiency will give us the extent to which the energy consumption differs from the expected consumption.

(ζmotordesign*ζpump design)/(ζmotor actual*ζpump actual) = Pactual (kW)/ Pdesign(kW)

0.85*0.65/0.77*0.45 = Pactual(kW)/ Pdesign(kW)

i.e Pactual (kW) is 1.6 times that of the expected Pdesign(kW)

Sl.No	Number hours of operation/day	Expected energy consumption/ month (assuming design efficiencies for motor and pump) (ζmotor -0.85,ζpump-0.65)	Actual energy consumption (average) /month
1	10	16200 units	22000 units
2	12	19400units	22000units
3	15	24400 units	22000 units

Table 7: Expected v/s actual energy consumption estimation

The scheme if operated for 15 hours a day for 30 days a month will consume units which are close to the average number of units being consumed by the scheme. But in reality, due to fluctuations in power and load shedding 15 hours/day throughout the month is not possible. Hence, the actual energy consumption is more than the expected.

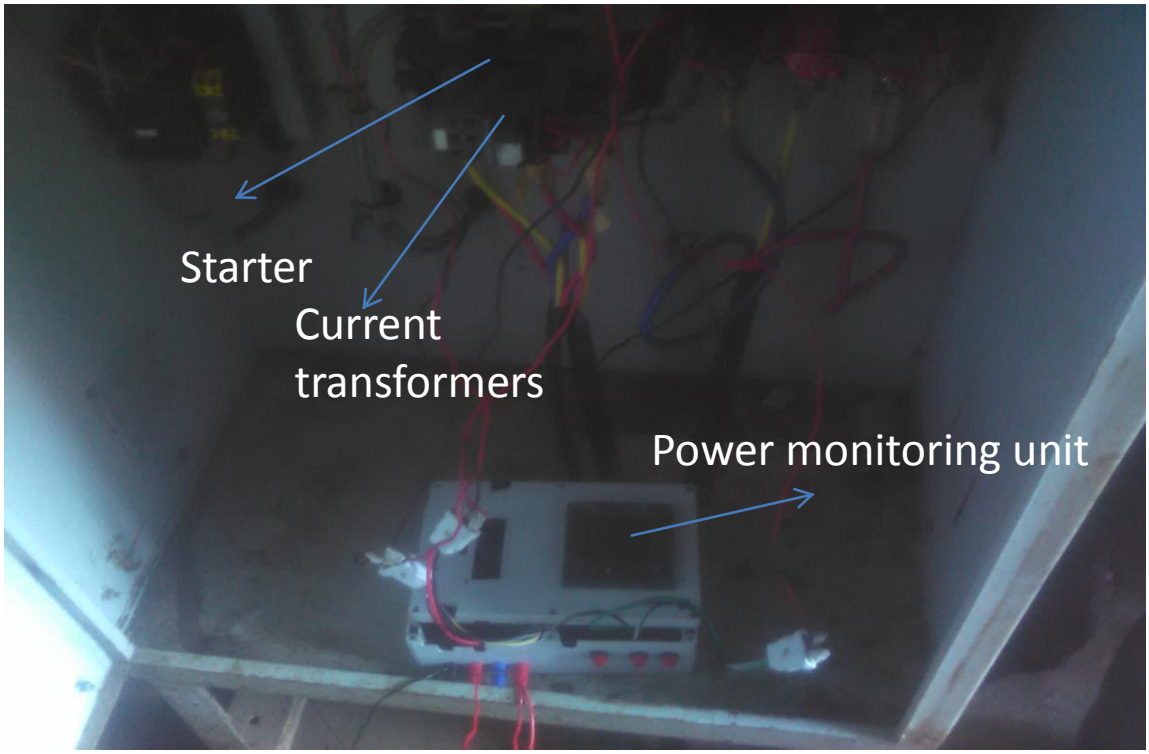


Figure 9 : Power monitoring unit and Current Transformer connections-Khardi

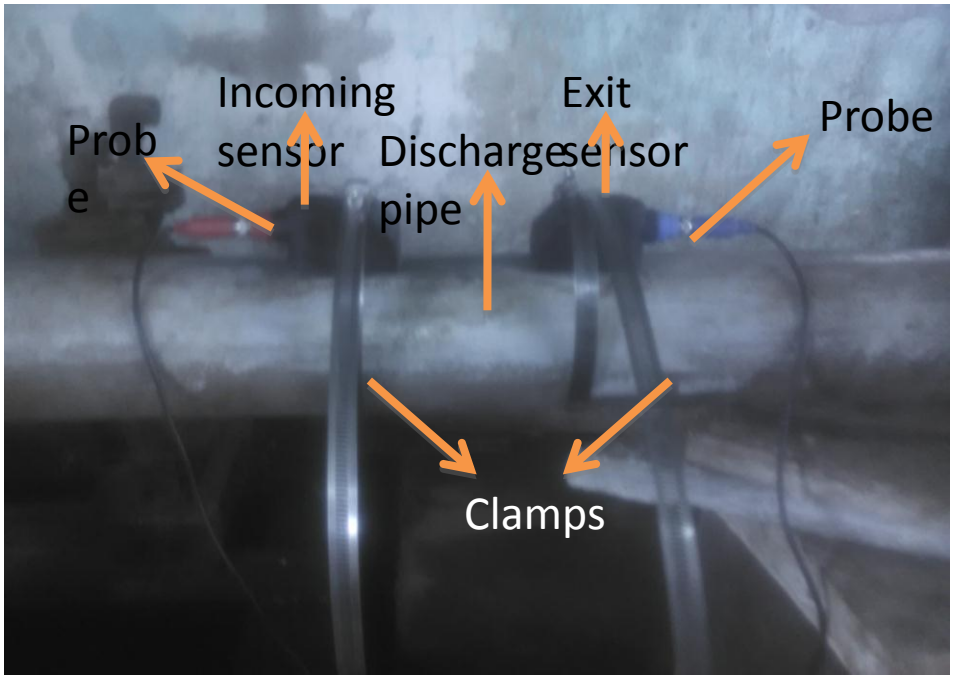


Figure 10: Measuring flow using ultrasonic sensors-Khardi

5.5. Water Loss and Leakage at Khardi

The water loss and leakages are the main source of inefficiency in rural piped water supply systems. There are mainly two types of losses the technical losses and non-technical losses. The technical losses are majorly due to damage in pipes, joints and fittings. The non-technical losses include theft; water wasted by consumer through open or leaky taps. Leaks increase the energy costs associated with pumping [7]. Khardi scheme has a 12 km long pipe line, which has a number of leakages some major and some minor. The leakage affects the efficiency of the scheme by reducing the pressure at which the water is supplied to the consumers and also increase in power consumption.

(i) Maintaining the demand according to the design

When equivalent service at demand points is maintained, the additional volume of water to be supplied due to leakage shifts the duty point of the system and lowers the head. The Brake Power needed increases with increase in flow rate and the frictional losses are proportional to the square of the flow rate. Thus the additional demand places a stress on the pump and hence increases the energy consumption and lowers the pressure at the delivery end.

The design is supposed to deliver the water at a design pressure and flow rate but the leakage affects the design duty point and lowers the pressure at which it is delivered. If the required pressure is to be maintained, the pump should be operated at a higher speed which needs higher Brake Power and more energy consumption.

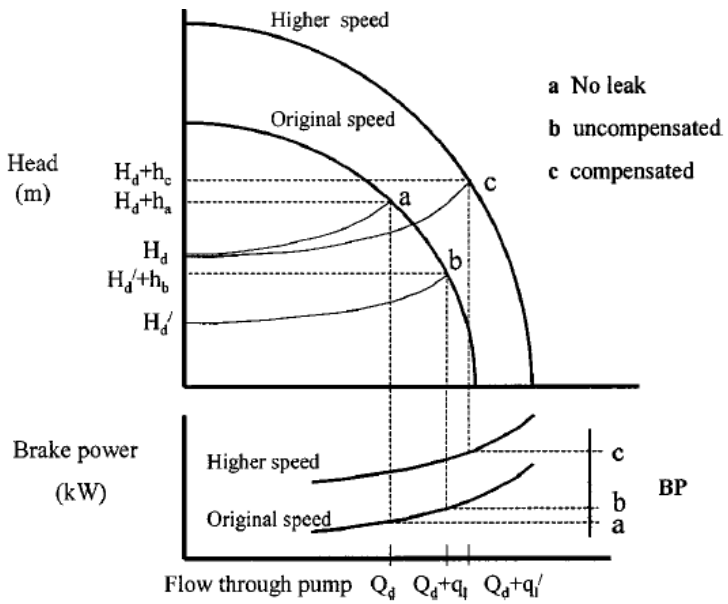


Figure 11: Pump curve showing the effect of leakage[6]

5.5Rahnvir Piped water supply scheme

Rahnvir is located in Shahpur taluka, about 23 km from Asangaon. The Piped water supply scheme is planned to supply water to Rahnvir pada, Madhilchiwadi, Lakhtachiwadi, Avlapada, Dholepada. There are two wells and two bore wells in the village. Two wells go dry in the summer and there is an acute shortage of water in the villages. So, the scheme is planned with source as a well near a stream in the outskirts of the village.

Table 8: Rahnvir PWS scheme details

Parameter	Value
Population targeted	1800
Design LPCD	40
BHP	25 H.P
Designed for number of hours of pumping	8 hours
Total design head	150 metres
Design flow rate	22 m^3/hour
Distance from Pump house to ESR	3 kilometres
Motor design efficiency (18.5 kW) – IS :9283 :2013	82%
Pump design efficiency for the specified head and flow rate according to pump curve	70%
Well depth	100 feet
Pump position	80 feet

Table 9: Measured Parameters-Rahnvir

Parameter	Value
V1,V2,V3	224V,217V,235V
Power factor	0.9
Average Flow rate	$24m^3$/hour
Average Pressure at pressure gauge	$11.2kg/cm^2$or 112 metres
Average Power input recorded	20.7kw
Water level from top	5 metres
Suction head	12 metres
Friction head upto pressure gauge with a total pipe length of 500 metres	0.8 metres

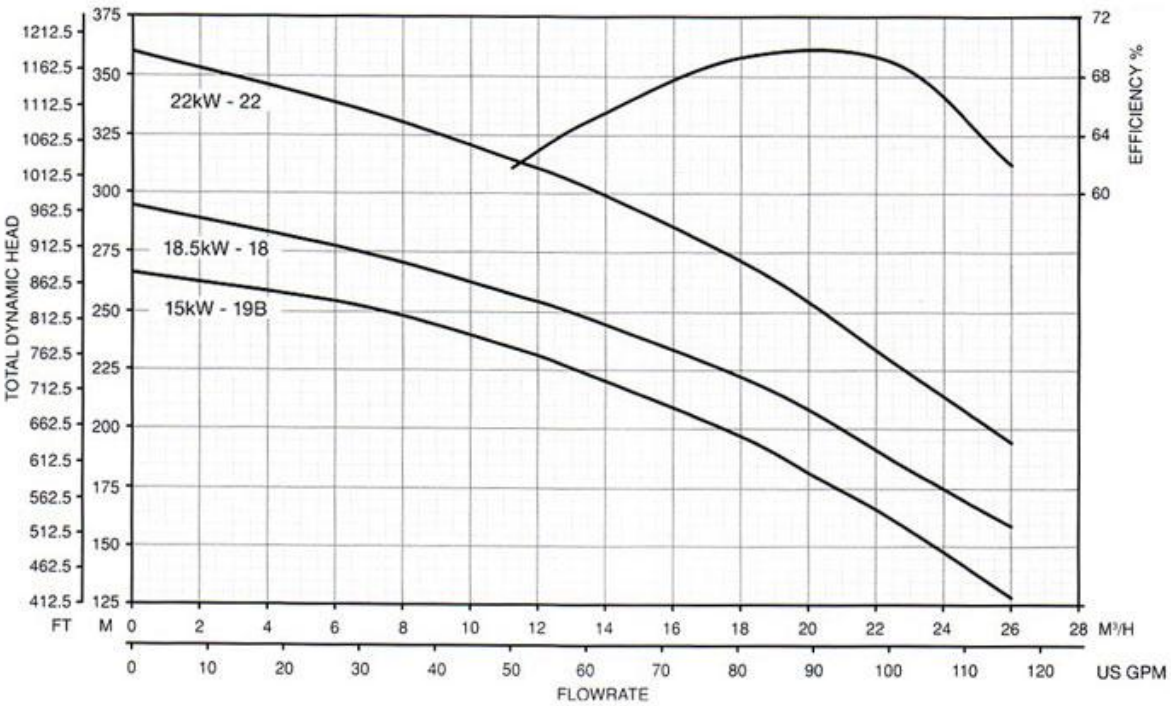


Figure 12: Pump characteristic curve for 18.5 kW, 18A submersible pump

5.6Efficiency calculation:

Total head as seen by the pump:

(Pressure at the pressure guage in meteres)+(Discharge head from pump to guage)+(friction head till the guage)-(suction head)

112m+24m+0.8m-12m=124.8

Hydraulic Power output: 7.3kW

Average power input recorded: 18.7kW

Total, Efficiency of the system: hydraulic output/power input = 40%

Pump efficiency for a head of 125m and flow rate of 24m^3/hour is 65%

Motor Efficiency is =Total efficiency/ Pump efficiency =61%

5.7 Comparing actual and design power consumption:

(ζmotordesign*ζpump design)/(ζmotor actual*ζpump actual) = Pactual (kW)/ Pdesign(kW)

0.82*0.70/0.61*0.65 = 1.44. That is actual power consumption is 1.44 times the expected Power consumption

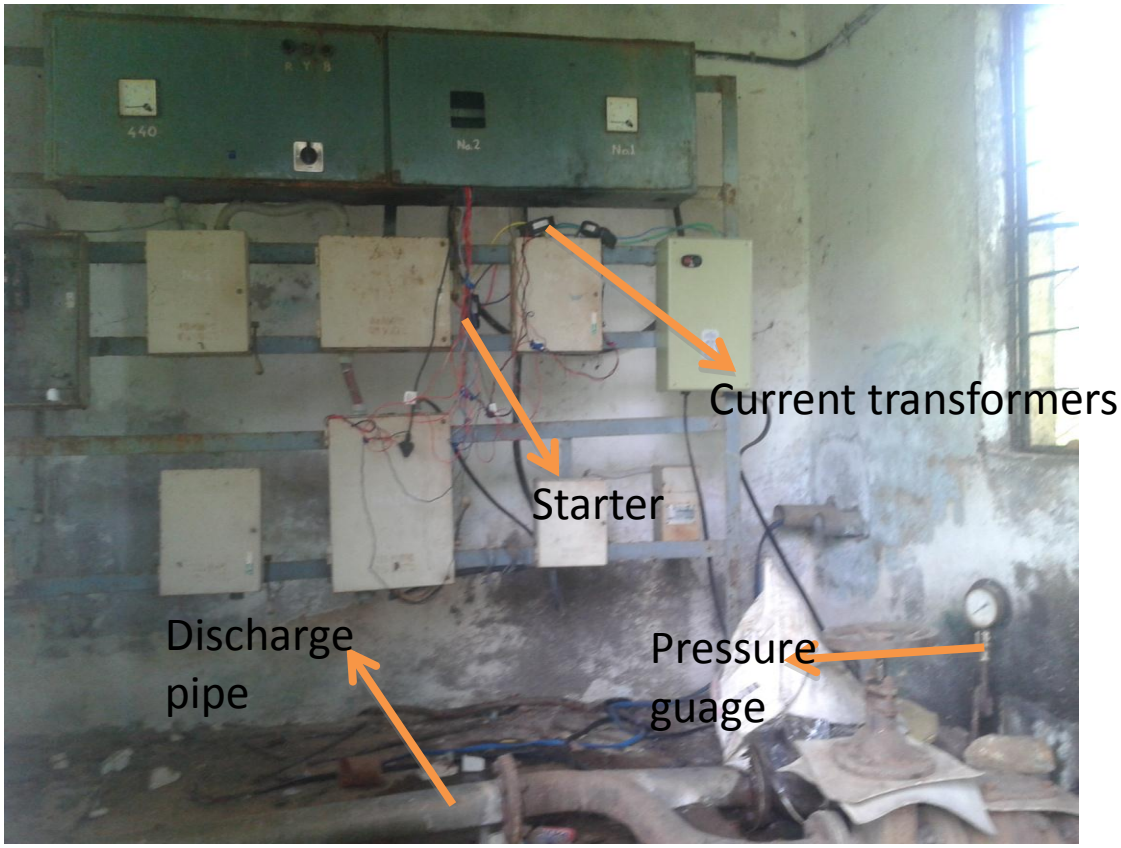


Figure 13: Rahnvir Pump house

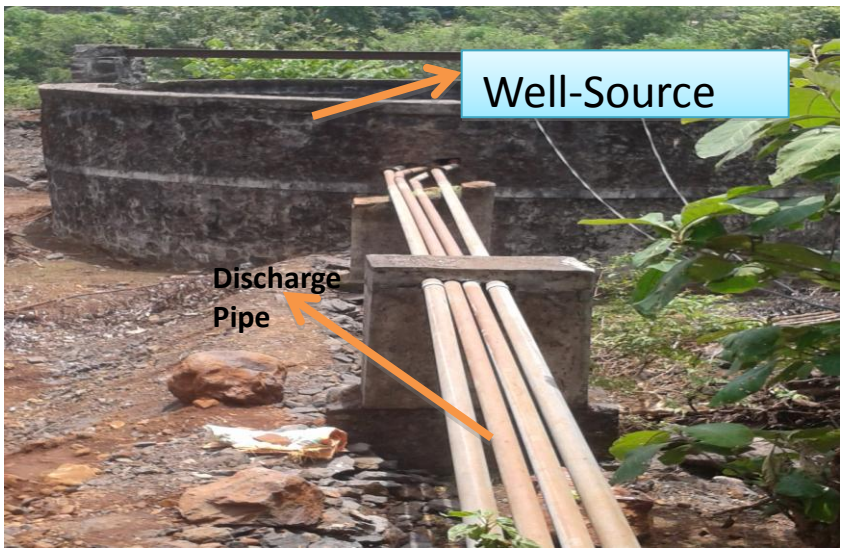


Figure 14: Source for Rahnvir scheme, the well and discharge pipes

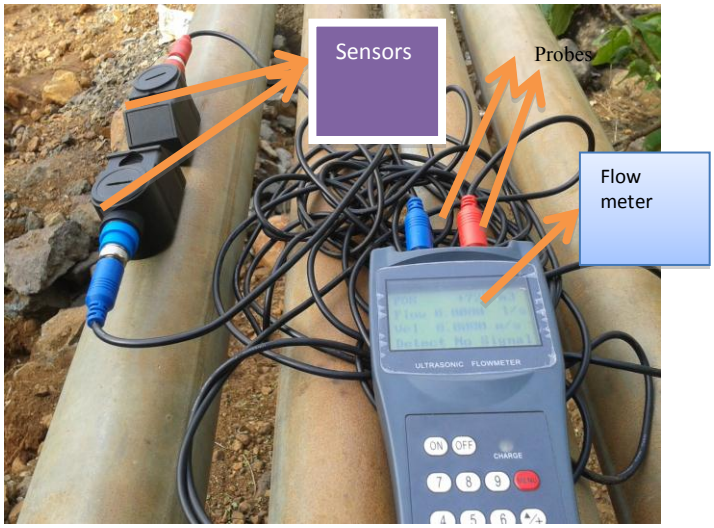


Figure 15: Flow measurement using ultrasonic flow meter

5.8 Limitations in testing the scheme for energy efficiency

(i) The major limitation in testing is finding the individual efficiencies of pump and motor unit separately. Most of the pumps are submersible and hence the motor efficiency could not be found.

(ii) Every pump is designed to operate at a particular speed, the pump performance curves are mentioned for a particular speed of operation. Since the pump is not accessible it is not possible to measure the speed, hence the pump characteristic curve used to determine the current efficiency level may not be accurate.

(iii) The suction determination may not be accurate as the exact placement of the pump inside the well is not known in most cases.

(iv) Availability of design documents is also an issue in many cases.

5.9 Issues faced during Testing

(i) The testing includes measurement of both power and flow rates. Public water supply systems are critical to people. So the testing has to be done without any interruption to the regular service, hence procuring the permission is mandatory.

(ii) Pursue the pump operators to cooperate with you in opening the MCC panel and fix the Power monitoring unit.

(iii) It is not possible to perform the measurements alone and at least two people are needed.

(iv) The frequent power outages in the supply schemes will create hurdles in measurement

(v) The Ultrasonic Flow Meter and Power Monitoring Unit may not work properly at the site due to varied reasons as there are different parameters and connections that have to be right. For example, the Ultrasonic Flow Meter may not detect the signal properly due to weak clamping. The Power Monitoring Unit may have faulty fuses or faulty internal wiring

7. Summary

This study aimed at understanding the current structure and status of PWS schemes and Literature review was done to understand the existing protocols underlined in the Detailed Design Document meant for Piped Water Supply schemes prepared by Ministry of Drinking water and Sanitation. Energy consumption of the PWS schemes was looked at in Shahpur Taluka and a sample template was prepared to assess the energy efficiency of schemes in general.Data measurements and calculations involving pump and motor efficiency were done so as to establish the efficiency of the schemes in Khardi and Rahnvir of Shahpur Taluka. Operation and maintenance protocols being followed, piping systems in use, effect of leakages on performance also have been studied.

7.1 Conclusion

- The Manual on Rural Drinking water supply systems, clearly underlines the protocols to be followed at each stage in execution of the scheme mentioning the type of precautions to be taken and systematic procedures to be followed at each stage of the scheme right from source selection, raw water mains inlet, pumping machinery, water treatment, leakage control and water audit, distribution, water billing and systematic systems management.
- In reality, the schemes in most cases do not take into account all the procedures laid down by the DPR guide lines as the work is executed by contractors and there is no supervision of technicalities so as to ensure the protocols are being followed strictly.
- The Piped Water Supply system's efficiency is typically a function of the motor pump and piping efficiencies, the motor and pump unit is designed to operate at the Best Efficiency Point, but during design phase, the possibility of geological changes due to draw down in the wells during summer, change in water level according to season change, leakages, encrustation in pipes leading to frictional losses are not taken care of properly leading to faulty systems.
- There is a certain level of disconnect between the Electricity and Water department in Maharashtraa's observed from the field study. The average time taken by the electricity department to respond to cases of under voltage or transformer burn out ranged from 2 days to 12 days as a result the O&M activities are not streamlined.
- The problem with the Piped Water Supply schemes is twofold - one is a very simplistic assumption about the loss in the pipe lines using a rule of thumb (general practice is to assume 10% losses) and the other is assumption of availability of power without any kind of agreement with MSEDCL or penalty structure in place.
- As a result of these assumptions, one tends to calculate a wrong flow rate. Also a partly knowledge about the suction head, change in friction head over a period of time results in improper selection of both head and flow rate, hence the design duty point varies from the actual duty point leading to inefficient operation.
- In a few cases, the pump might match properly, but the motor coupled with the pump may be overloaded or inefficient due to increase in winding temperature or slip losses.
- It is a good idea to install bulk water meters across the line to check leakages at least in schemes with higher capacity to justify the cost and improve the O&M practice. Also, a phased replacement of leaky pipe lines is necessary.

References:

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2. Manual for Preparation of Detailed Project Report– Ministry of Drinking Water and Sanitation, Government of India, February, 2013.

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4. Evaluation of water pumping systems -Energy efficiency manual Page 32

5. Frictional loss in pipe lines-http://www.engineeringtoolbox.com/hazen-williams

6. IS14220 (1994): Code for acceptance of Open submersible Pump sets, Govt of India

7. IS 9283 (2013): Code for acceptance test for motor efficiency ,Govt of India

8. Pump parameters –www.webpump.com

9. Impacts of Leaks on Energy Consumption in Pumped Systems with Storage, Andrew F. Colombo and Bryan W. Karney, M.ASCE, April 2005

Appendix A

Ultrasonic Flow meter Usage

The Ultrasonic flow meter can be used to measure the flow/velocity of the water in the pipe lines. It has two Ultrasonic sensors, probes and measuring device. Though the operation is very easy and the manual clearly mentions the usage, the following are the few things that one has to keep in mind before using it for any measurement.

(i) The instrument is factory calibrated and does not need recalibration as it is robust if need be, the calibration can be tested in hydraulics lab with another Ultrasonic flow meter working parallel to measure the flow of the liquid and see if both show the same reading.

(ii) The meter might not detect signal if not clamped well to the pipe, hence see to it that the pipe's outer surface is clean without any dust, paint or rust by scrapping it with a sand paper.

(iii) Apply electrode gel to the sensors if available alternatively you can also apply Vaseline in small amounts so as to create transparency and better gripping capability to the pipe.

(iv) Clamp the sensors with the help of the clampers by measuring the distance between them as indicated by the meter. The distance of separation depends upon the type of clamping (V method, Z method), pipe diameter, and pipe thickness. Once these parameters are entered the distance that needs to be maintained is automatically indicated by the meter. Follow Z method clamping as indicated in the manual.

(v) Always use the flow meter with at least 10 times the diameter's distance from the bends on both sides because the bends change the nature of the flow and may result in erroneous observation.

Figure: Ultrasonic flow meter calibration at hydraulics lab

Appendix B

Format for data collection to calculate energy efficiency

Parameter	Value
Design head	
Design flow rate	
Level at which pump is placed inside the sump	
Total depth of the well	
Design efficiency from pump curve	
Water level from the top	

Parameter	Value
Pressure at the pressure gauge	
Discharge head till the gauge	
Friction head till the gauge	
Suction head	
Measured flow rate	
Pump efficiency for the given duty point	

Power readings format:

V1	V2	V3	I1	I2	I3	P(kW)	frequency	Power factor